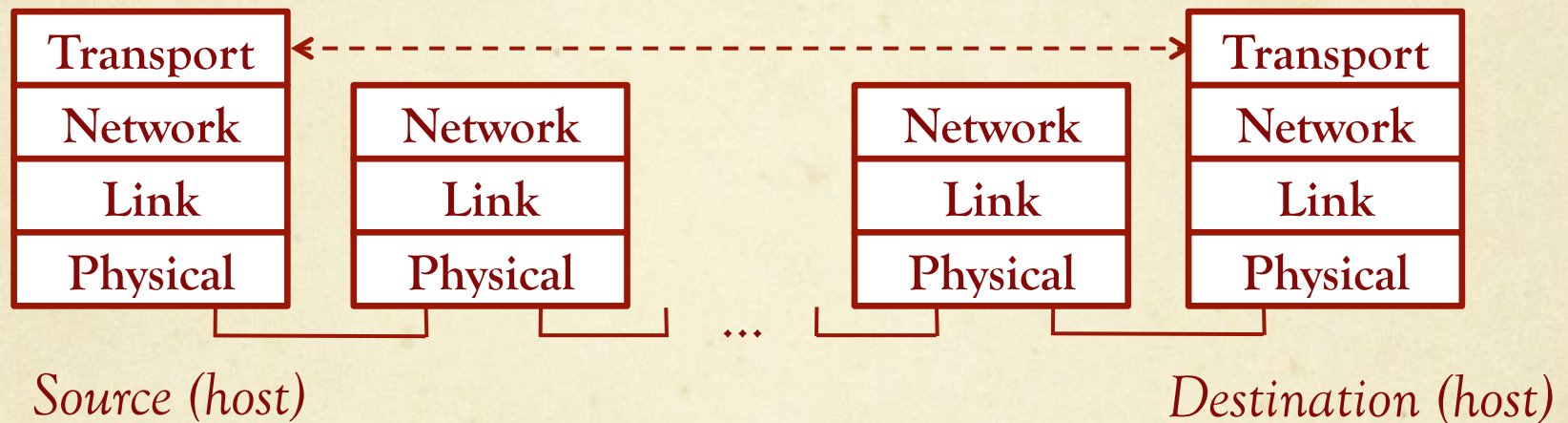


The Transport Layer

Transport layer

- ◆ Provide *logical* end-to-end communication between source & destination



- ◆ Run entirely on *host* computers
 - ◆ Network layer protocols run on *hosts* & *routers*

Transport layer

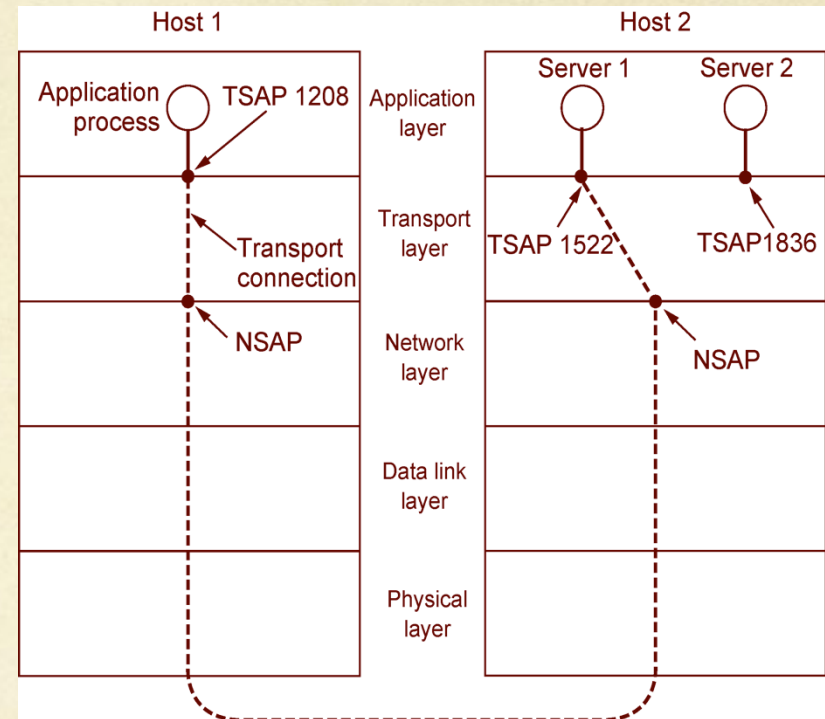
- ◆ Supports concurrent network usage by *multiple* processes
 - ◆ Provides comm. b/w *application processes* on hosts
 - ◆ Provides clean *interface* for processes to use
- ◆ *Relies on & enhance* network layer services

Transport layer

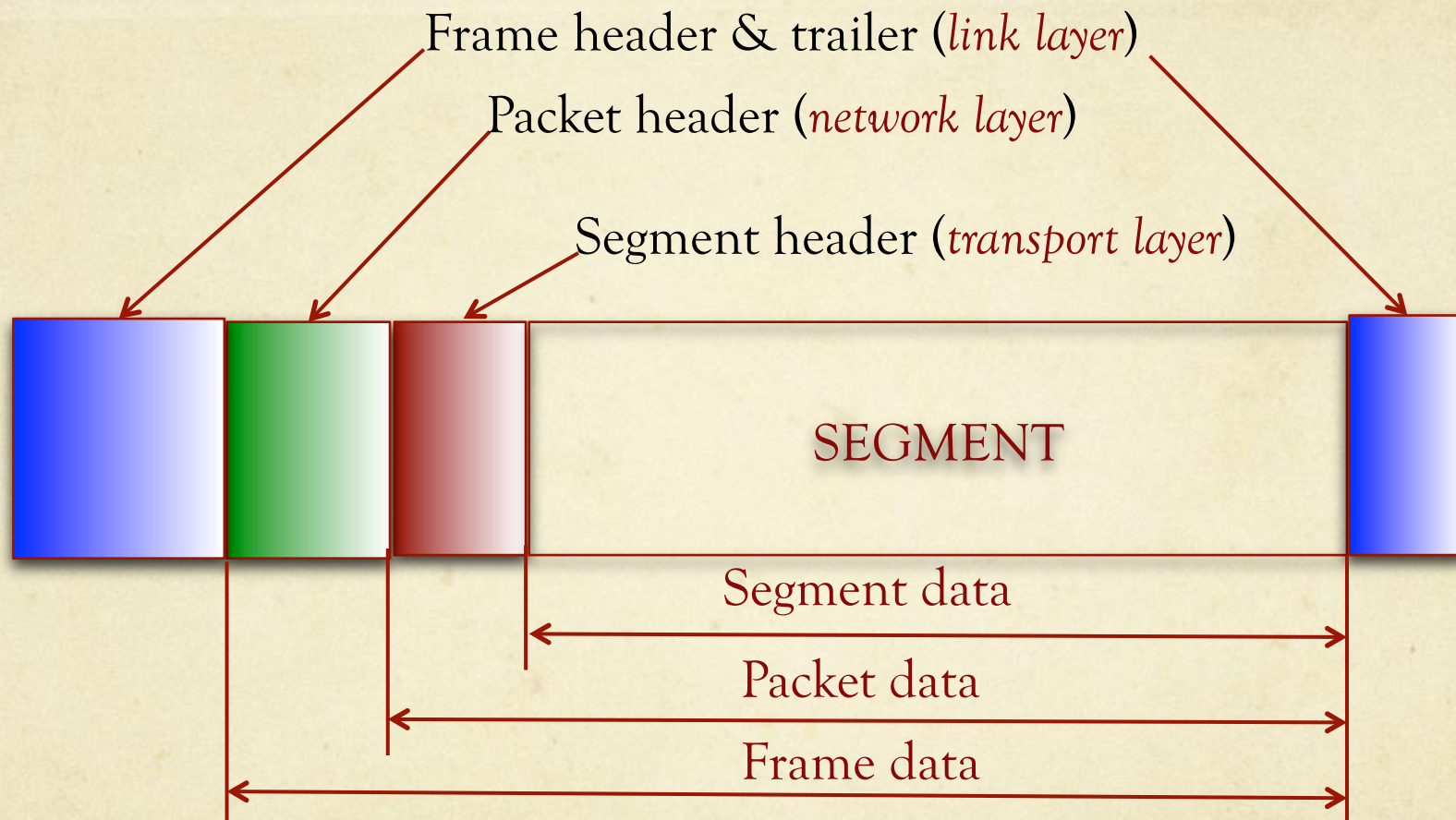
- ◆ Can provide *connection-oriented* or *connectionless* service
- ◆ Can provide following features
 - ◆ Reliability [*even on top of an unreliable network*]
 - ◆ Flow control
 - ◆ Congestion control

Supporting multiple processes

- ◆ Know how to identify *hosts*
 - ◆ IP address
 - ◆ Network Service Access Point (*NSAP*)
- ◆ Need to identify *processes* in host
 - ◆ *Port* number
 - ◆ Transport Service Access Point (*TSAP*)
- ◆ Host address + process port number → *socket*



Let's clarify some terminology...



Transport layer protocols in the *Internet*

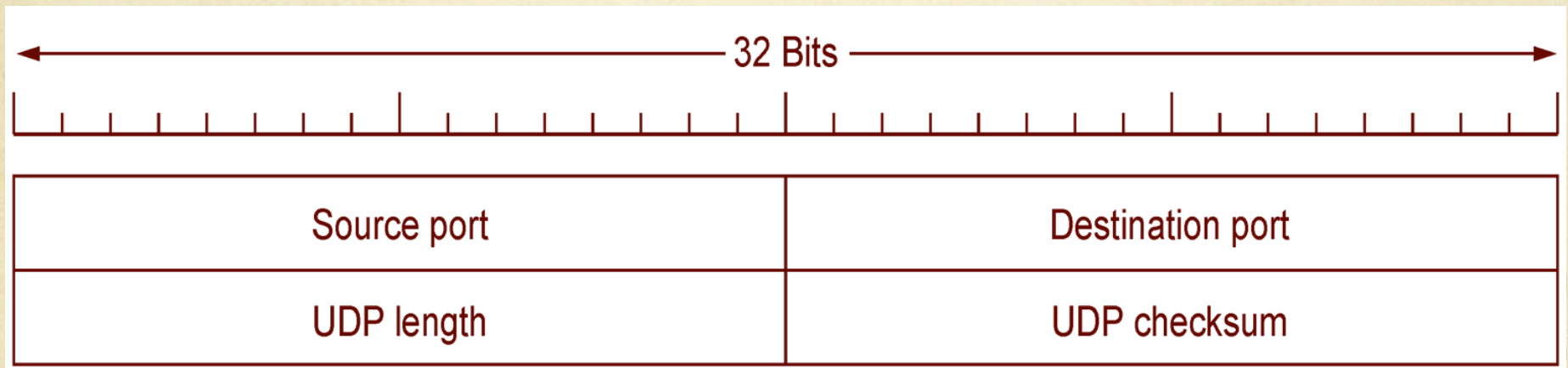
- ◆ We will look at two widely used protocols
 - ◆ User Datagram Protocol (*UDP*)
 - ◆ Transmission Control Protocol (*TCP*)

User Datagram Protocol (*UDP*)

- ◆ *Unreliable* (best-effort)
- ◆ *Connectionless*
- ◆ Limited error control
- ◆ No flow control or congestion control
- ◆ Suitable for
 - ◆ Request-response communication with no error/flow control
 - ◆ Use with applications that handle error/flow control
 - ◆ *Multicast* or *broadcast* communication

User Datagram Protocol (*UDP*)

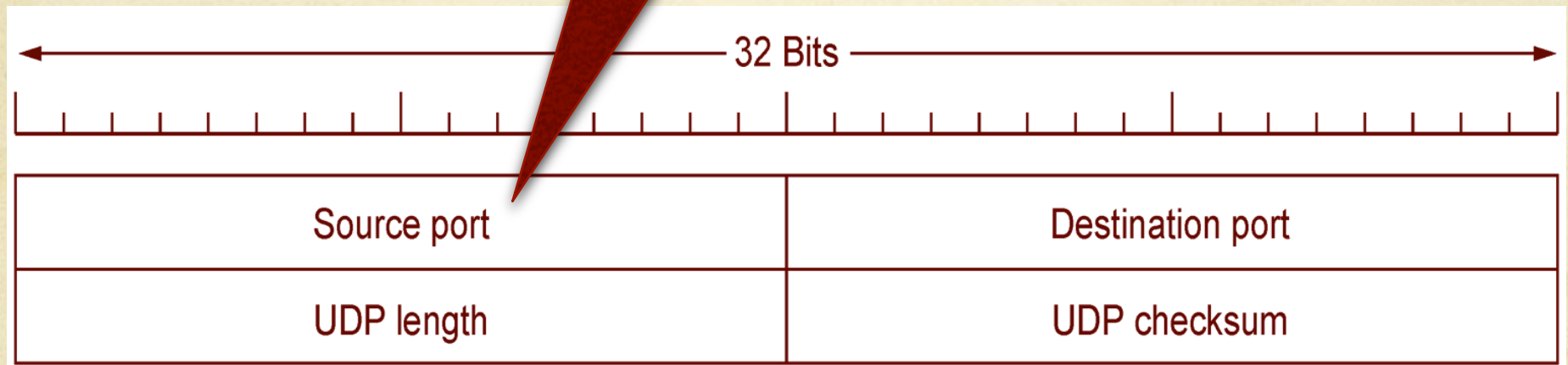
- ◆ UDP *header*



User Datagram Protocol (*UDP*)

- ◆ UDP *header*

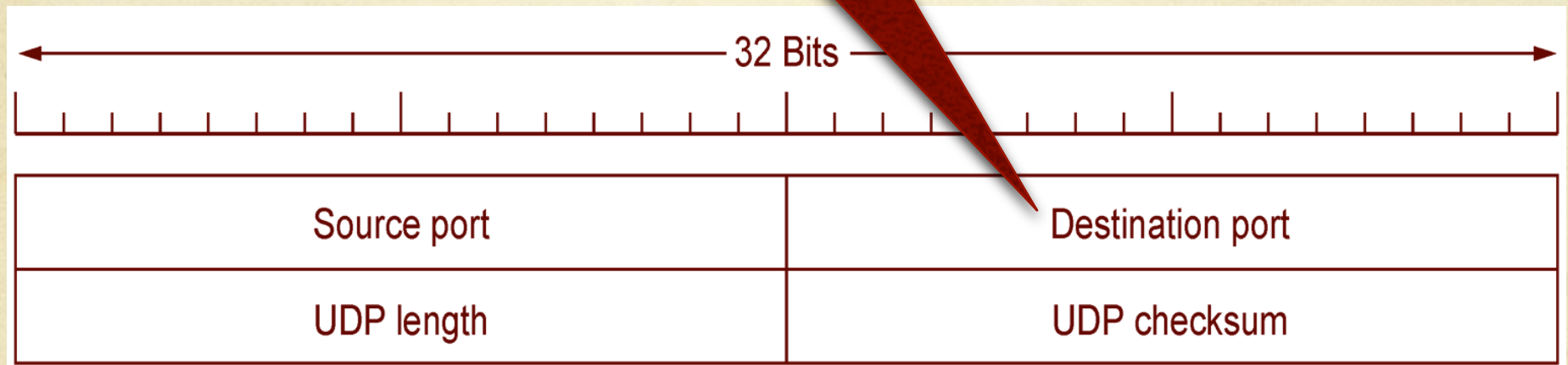
*Application port
number within source
host*



User Datagram Protocol (*UDP*)

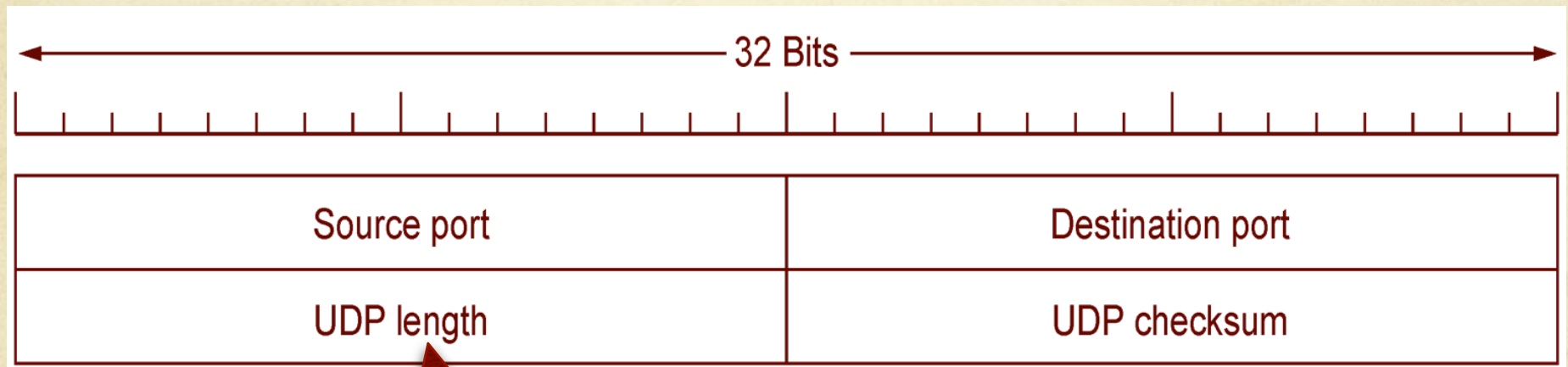
- ◆ UDP *header*

*Application port
number within
destination host*



User Datagram Protocol (*UDP*)

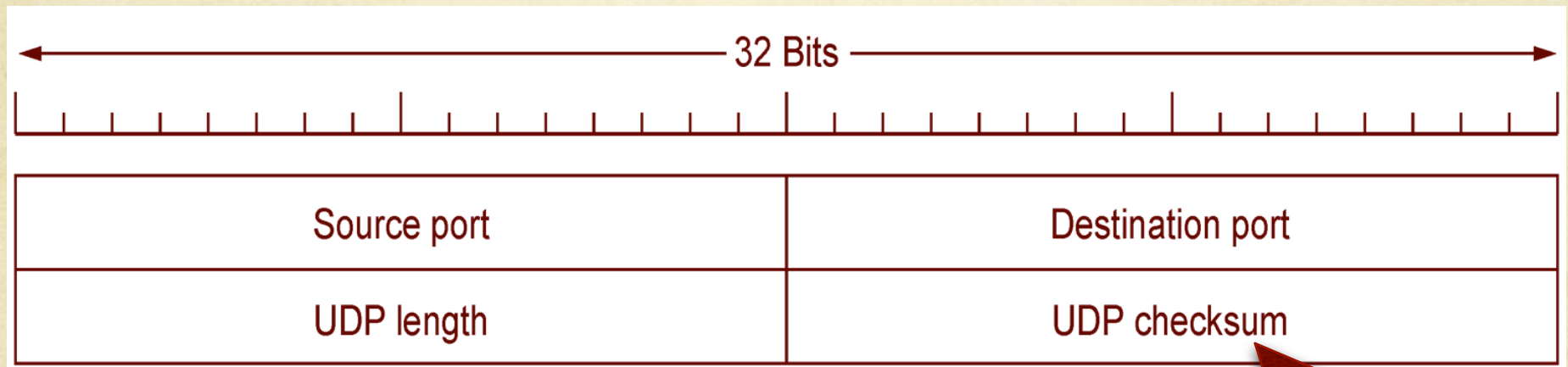
- ◆ UDP *header*



8 bytes header
length + length of
data

User Datagram Protocol (*UDP*)

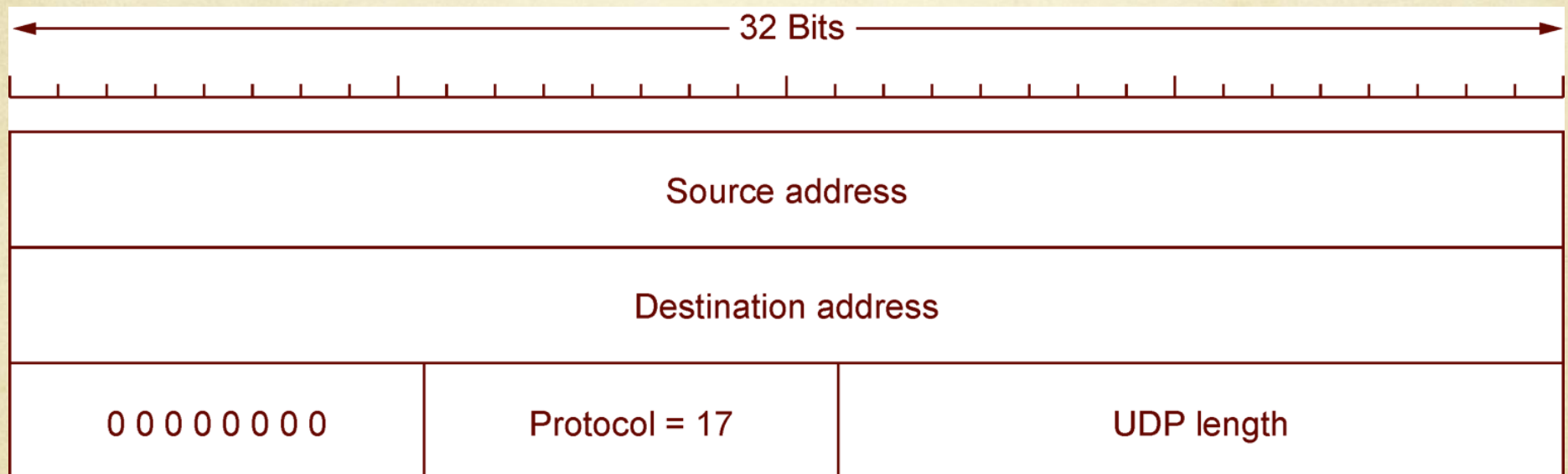
- ◆ UDP *header*



Optional checksum for UDP
header, IPv4 pseudo-header
& data

User Datagram Protocol (*UDP*)

- ◆ IPv4 *pseudo-header* included in UDP checksum



User Datagram Protocol (*UDP*)

- ◆ IPv4 *pseudo-header* included in UDP checksum
 - ◆ *Strictly speaking, this violates layering*

